

ISMG 4860/6860: Ethical Hacking Concepts and Methodologies

Fall 2025

Instructor: Dr. Ersin Dincelli

Email: ersin.dincelli@ucdenver.edu

Office: BUSB 5102

Phone: 303-315-8489

Class Location: Business School Building 2501*

Course Format: 8-week, Hybrid*/Online (asynchronous)

TA: Lukasz Kusyk (lukasz.kusyk@ucdenver.edu)

Office Hours: Wednesday, 12 pm-1 pm, online (see below)

*Class Meetings (optional): 8/20, 8/27, 9/10, 9/17, 9/24, 10/8 (at 6:30 pm)

Course Description:

From a technical perspective, organizations need to know how hackers work so that they can build their security around it and take preemptive measures against future attacks. The goal of ethical hacking is to understand current exploits and assess weaknesses and vulnerabilities of various organizational information systems by attacking them within legal limits. This course is designed to provide students with an insight into current hacking tools and techniques used by hackers and security professionals to break into any computer systems. Throughout the course, students will engage in offensive and defensive hands-on exercises stressing ethical hacking and penetration testing that will be conducted in a vendor-neutral virtual environment. Topics include security threats and attack vectors, footprinting and reconnaissance, Google hacking, social engineering, insider threat, network scanning and enumeration techniques, vulnerability assessment, the Dark Web, and attack and defense strategies in emerging technologies, such as the Internet of Things (IoT) and cloud computing.

Course Prerequisites:

A basic understanding of operating systems and the Internet is required. Course content, assignments, exams, and announcements will be posted on Canvas. Therefore, access to the Internet and a personal computer (Windows or macOS are recommended) are required to successfully complete this course. Online exams will be proctored, requiring a working webcam and proctoring software. It is not required, but I recommend taking ISMG 6120 – Network Design and Analysis as a complementary course to enhance your understanding of cybersecurity.

Course Learning Goals and Objectives:

Students who successfully complete this class will be able to: (1) Identify the legal and ethical implications of hacking. (2) Compare and contrast different forms of hacking activities and techniques. (3) Evaluate different vulnerabilities, threats, and attacks to information systems and recommend countermeasures. (4) Apply various network security assessments and hacking tools to real-world scenarios in a secure virtual environment. (5) Implement the appropriate level of information security controls based on evidence, information, and research. This course contributes to the following IS program level learning outcomes:

- **Goal 4:** Act Securely; Objective: Apply basic cybersecurity principles to identify, protect, detect, respond and recover from cybersecurity threats.
- **Goal 5:** Behave Ethically; Objective: Assess ethical issues in information systems, such as privacy, access and bias

Course Materials:

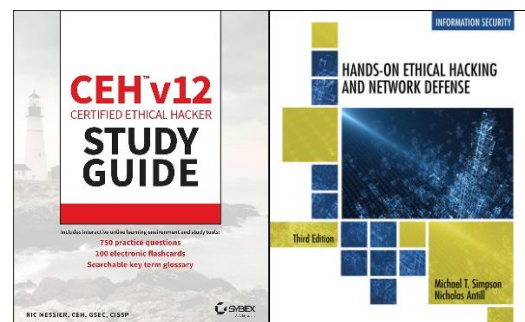
Required Textbook: *CEH v12 Certified Ethical Hacker Study Guide*
1st Ed. By Ric Messier. ISBN-10: 1394186924.

[\[Wiley\]](#) [\[Amazon\]](#)

* You might also consider buying the study guide for CEH v10 or v11, as they may be cheaper. The differences between the older and newer versions are minimal.

Optional: *Hands-On Ethical Hacking and Network Defense* 3rd Ed.
By Michael T. Simpson and Nicholas Antill. ISBN-10: 1285454618.

[\[Cengage\]](#) [\[Amazon\]](#)



Virtual Labs: CEH v12 Volume 1: Ethical Hacking Concepts and Methodology

You can buy the iLabs on the discounted academia partner eStore as it is significantly cheaper than other alternatives:

- Go to [Academia Partner eStore](#)
- Click "Accept," and then click "Proceed to checkout"
- Enter the requested info, click "Ship to this address," then "Continue"
- Enter the school name: "University of Colorado Denver"
- Click "Continue to Payment," enter payment details, confirm the order



Throughout the semester, we are going to be using the first volume of **CEHv12** iLabs, which includes hands-on activities on *footprinting and reconnaissance*, *scanning networks*, *enumeration*, and *vulnerability analysis* in a secure virtual environment. **Please note that you only need to buy the iLabs volume 1 for \$49 for this course.**

Course Meetings:

This course offers both hybrid and online sections. Hybrid courses combine elements of traditional and online classes. To provide elements of traditional classes, this class meets six times based on the schedule below. These meetings will also be streamed live via Zoom. Attending in-person meetings is optional. Lecture recordings will be posted online to ensure that online students have access to all the content.

Tentative Course Schedule*:

Unit/Date	Topic	Readings	Assignments & Deliverables	Due
Unit 1 8/18-8/24	Class Meeting (8/20) Introduction to Ethical Hacking Security Foundations	Ch. 1, 3	Introduction posts Online Discussion #1: Hacktivism	8/24 8/24
Unit 2 8/25-8/31	Footprinting and Reconnaissance Google Hacking, Class Meeting (8/27) iLabs Exercises #1	Ch. 4	Online Discussion #1 responses Assignment #1: Reconnaissance	8/31 8/31
Unit 3 9/1-9/7	Networking Foundations Scanning Networks iLabs Exercises #2 (No class meeting)	Ch. 2, 5	Online Discussion #2: Missing People Assignment #2: Scanning Networks	9/7 9/7
Unit 4 9/8-9/14	Class Meeting (9/10) Social Engineering & Insider Threat Online Midterm Exam on 9/12	Ch. 10	Project Proposal (Graduate) Online Discussion #2 responses	9/14 9/14
9/10	Guest Speaker: Chris Edmundson, Security Operations Manager, University of Colorado Denver			
Unit 5 9/15-9/21	Class Meeting (9/17) Enumeration iLabs Exercises #3	Ch. 6	Online Discussion #3: Insiders Assignment #3: Enumeration	9/21 9/21
Unit 6 9/22-9/28	Class Meeting (9/24) Vulnerability Analysis, Dark Web iLabs exercises #4	Notes	Online Discussion #3 responses Assignment #4: Vulnerability Analysis	9/28 9/28
Unit 7 9/29-10/5	Sniffing (No class meeting) Cloud Computing Security	Ch. 9	Online Discussion #4: AI & Privacy	10/5
Unit 8 10/6-10/11	Class Meeting (10/8) IoT Security, Final Exam Review Online Final Exam on 10/11	Ch. 15, Notes	Online Discussion #4 responses Project Write-up (Graduate)	10/10 10/11
10/8	Guest Speaker: Ryan Johnson, Senior Partner at DigiStream Intelligence Tradecraft: Principles for Advanced Investigations			

*Topics and the order of the topics may be subject to change.

We are going to have six in-class meetings. These weeks are marked with green.

We are not going to have class meetings in weeks 3 and 7. They are marked with red.

Course Assignments and Grading:

(1) Assignments: You have to complete all the assignments **as an individual, not with other students.**

Copying others' assignments or purchasing answers from websites that monetize course materials will be considered cheating. All assignments will be available on Canvas. You are required to complete the assigned readings, iLabs, and deliverables before completing each assignment. The due date for completing each assignment is specified in the course schedule and Canvas. **No late assignments will be accepted.** I grade assignments as follows:

Points	Justification
17-20	You have done an outstanding job!
11-16	You have clearly completed the assigned chapters and other deliverables and made an honest attempt to apply it to the assignment.
6-10	You have not been particularly successful in completing all the requirements of the assignment.
1-5	You have completely missed the point of the assignment, or your answer shows little evidence of real effort.
0	You did not hand in the assignment on time. Not submitting assignments can add up to be costly, usually the difference in a letter grade.

If you have clearly read the assigned chapters, complete the labs, and made an honest attempt to apply your knowledge to the assignment, you will receive a minimum of 16 points. I expect almost everyone to get 19+.

(2) Course Project (only graduate students need to complete the course project): Throughout the semester, you will be exposed to various security-related topics. For the course project, you will identify a topic of your interest and write a **short paragraph as a proposal**. Then, you will write a **four-page** research paper or a report about a topic (e.g., introducing a tool that tackles a security problem, innovative security education and training programs, etc.) I will provide feedback based on your proposal. You have the option to submit your paper to receive further feedback from me throughout the semester. Please check the project assignment space on Canvas for the formatting requirements and the grading rubric. Please see this project as an opportunity to work on something that you are interested in. I am flexible in terms of topics. Feel free to get in touch with me any time you want to discuss your project.

(3) Midterm Exam: The midterm exam is a closed book, online exam and will assess your understanding (as an individual, not as a team) of course material. The exam will be proctored online via [Respondus Lockdown Browser](#). I highly suggest you avoid taking it at the same time with your roommates and using the same Internet connection to avoid potential red flags. If you prefer, you may take the exam on campus using a school computer. The exam will consist of 35 multiple-choice and true/false questions, covering all materials up to the exam date.

(4) Final Exam: The format of the final exam is the same as the midterm exam (closed book, online, and proctored). The final exam will cover all the materials covered until the final exam date.

(5) Online Discussions: The purpose of the online discussions is to learn from personal knowledge and/or experiences of each other. Active participation in the online discussion is an important component of this course. It is expected that you enrich the course through relevant discussions and contributions of personal knowledge and experience. You should read over comments made by others and respond to the comments. I encourage you to add value to the discussion by bringing a new perspective to what was said rather than simply repeating the points. Examples of acceptable discussion material are (1) your own interpretations of issues discussed in the readings, (2) relating conceptual material from the readings to real-world examples, (3) drawing parallels with or integrating ideas covered in previous classes, (4) critiquing a classmate's analysis, (5) applying outside readings to the discussion at hand, and (6) demonstrating that you have carefully read the material and thought about it. Evaluation of your participation will be based on substantive contribution to the class or group learning experience, and not merely on how many times you entered your comments.

In a typical online discussion, I will pose a topic for discussion or a series of topics, and you, as the student, can respond to the topics and/or respond to the other students' responses. You enter the online discussion by clicking on the respond button on the thread page. Your participation will be evaluated as follows:

Points	Rubric for Online Content Assessment	Rubric for Assessing Online Participation
17-20	Demonstrates excellence in grasping key concepts; critiques others' work; provides ample evidence of support for opinions; readily offers new interpretations of discussion material.	Contributions are prompt, timely, relevant, self-initiated; remarks are posted freely on all assignments throughout the course; there is no attempt to dominate the conversation.
11-16	Shows evidence of understanding most of the major concepts; is able to agree or disagree when prompted; is skilled in the basic level of support for opinions; offers an occasional divergent viewpoint.	The student generally keeps up with the discussion; needs occasional prompting to contribute; might participate in some discussions more than others.
4-10	Has a mostly shallow grasp of the material; rarely takes a stand on issues; offers inadequate levels of support.	Participation is spotty; picks and chooses topics to get involved in; offers short, perfunctory postings when prompted; takes limited initiative.
1-3	Shows no significant understanding of the material.	The student rarely participates freely; makes short, irrelevant remarks.

Remember, you need to discuss the questions in your group rather than just answering the questions. It is important to participate early (rather than the last day) and very often to see everybody's contributions in order to be able to add value to the discussion. In your discussions and especially in the open-ended questions, you need to (1) make inferences, (2) relate new knowledge to prior knowledge and experience, and (3) interpret content through the analysis, synthesis, and evaluation of others' understanding. The highest grade for the online participation portion of the online discussions is achieved **(1) if you post at least four responses (your initial post does not count toward the four) during the discussion period** (you can respond to the same or different students), **(2) add value to discussion each time**, and **(3) spread your responses over time**. Waiting until the last day of the discussions to post your responses will be penalized. Late online discussions will NOT be accepted. Each individual thread worth **20 points (10 from your initial response and 10 from your responses)**.

Grading Policy: Your semester grade will be based on the following criteria:

Deliverables	Percentage	
	Undergraduate	Graduate
Midterm Exam	30%	25%
Final Exam	30%	25%
Assignments	20%	20%
Discussions	20%	20%
Course Project	n/a	10%

Letter Grades Scale			
95 – 100%	A	73 – 76%	C
90 – 94%	A-	70 – 72%	C-
87 – 89%	B+	67 – 69%	D+
83 – 86%	B	63 – 66%	D
80 – 82%	B-	60 – 62%	D-
77 – 79%	C+		

Grades are not rounded. For example, 89.9% is B+. At the discretion of the instructor, grades will be "curved." However, no student will receive a grade less than that prescribed by the above schedule. **There will be absolutely no extensions to any deadlines.** Completing assignments on time is critical to your success in the course! Attendance at examinations is mandatory. Makeup exams are not typically given. The only acceptable excuses for missing an exam are circumstances clearly outside your control, such as illness, death in the family, etc. If you miss an exam, notify me as soon as possible. I will require documentation of the circumstances.

Office Hours:

We will have an online office hour every week. Meetings will be held through Zoom. Please check Canvas for the Zoom link and other details. To use the time efficiently, you need to send our TA an email, including the questions

you want to ask during office hours. We can also schedule a meeting if you cannot make it to the office hours. We will get back to you within 24 hours. Our email addresses are listed on the first page of the syllabus.

Course Approach:

This is an 8-week long class. This means **you need to spend two times more time than a regular 16-week long class** each week. You should expect to spend about 12 to 18 hours per unit on this course. If you are working full-time, you might struggle to keep up with the required work. All lectures and course materials will be available online. You need to regularly log in to the course to check announcements, participate in online discussions, and access course content. There are assignments and online discussions for each unit. You are expected to participate in the online discussions by the due dates. You need to read the weekly objectives and checklist on the week's homepage. This gives an overview of activities to be completed in the coming week. Go to the Q&A page several times throughout the week. I will answer/follow-up on questions that are discussed during the week and give hints about the assignments there. Exams will be based on the textbook, readings, online discussions, slides, and labs. The material covered in each unit closely follows the course text and the assigned readings. See the calendar and units for details. We will cover one unit per week.

This course reflects a learning-centered, rather than an instruction-centered, paradigm. Whereas the instruction-centered paradigm views learning as an essentially passive process of assimilating "downloaded" information transmitted by an instructor, the learning-centered paradigm recognizes that learning involves the construction by the learner. Some of the assumptions and implications of following a learning-centered paradigm are:

- The course aims at producing learning rather than merely providing or delivering knowledge
- Student discovery and construction of knowledge are emphasized over the transfer of knowledge from faculty to students
- Knowledge exists in the mind of the student and is shaped by the student's experience; knowledge is not purely "objective"
- Learning is student-centered and controlled
- Learning is cooperative, collaborative, and supportive rather than competitive
- Talent and ability are seen as abundant among the students
- Faculty are primarily designers of learning methods and environments

Succeeding Online:

Learning online is both a great educational opportunity and a challenge. Although you can better schedule your course time around your needs, you also must take greater responsibility for managing your time and getting work done. You will find that not having a class means you have to do more written work since you cannot receive grades for class participation or attendance. This course is structured with work and graded assignments due every week. The best way to succeed in this environment is to plan on doing weekly work to stay on top of the course load and don't fall behind. You should expect to spend about 18 hours per unit on this course as it is an 8-week long course. This includes the time you will need to complete the readings, as well as online discussion work that week. Do not expect to succeed in this course if you do not allocate at least 18 hours a week! With each new unit and assignment, you build on the work that was completed so far from previous weeks. Therefore, it is crucial to complete weekly assignments on time. Students who fall behind may find it very difficult to catch up. The best approach is to schedule the time and keep up with work. Online classes promise to provide an excellent educational opportunity. But the ultimate success of the course sits on your shoulders.

Etiquette and Conduct:

As a University of Colorado Denver student (or taking a UCD class), you are expected to abide by the University of Colorado Denver [Student Code of Conduct](#).

Academic Honesty (PLEASE READ CAREFULLY):

When you log in to your courses using your CU Denver login credentials, you are certifying that the person logging in is you. As such, providing your credentials to any other individuals and **allowing them to log in to a course on**

your behalf is considered academic dishonesty. You are expected to **do your own work**. All work submitted should include citations or other indications when other's work is included with your own. **Representation of others' work as your own is considered cheating**. This includes plagiarism, **answers directly copied from "ChatGPT"**, **work purchased from "paper mills,"** and **sharing of test answers** as well as **sharing your work with other students or websites that monetize assignments**. Any time you put your name on a piece of work for this course, you assert that it is your own work, except as otherwise indicated and permitted. If that is not the case, it is academic dishonesty, and **I am required to submit the matter to the College Internal Affairs committee for a decision**. Finally, cheating diminishes the value of your learning. If you find yourself struggling with this course, **please contact the instructor!**

IW/IF Policy:

According to college policy, in order to receive an IF or IW the following five items must have been approved:

- The student must have completed at least 75% of the course with a passing grade, and there must be compelling, extraordinary verifiable circumstances beyond the student's control, which made the completion of the course impossible.
- The student must submit a copy of the syllabus and verification of reason (e.g., doctor's note, letter the dean explaining the situation, etc.).
- The form must be signed by faculty and have attached documents before being submitted to the Associate Dean.
- Paperwork must be processed and approved by the dean before grades are assigned.
- The student has one academic year to complete the work, as noted by the instructor.

Disability Services Information:

To ensure disability-related concerns are properly addressed, students with disabilities who require assistance to participate in this class are expected to contact me. For further information regarding disability-related accommodations, contact the Office of Disability Resources and Services, 177 Arts Building at 303.556.3450.

Family Educational Rights and Privacy Act (FERPA):

FERPA was created to protect the privacy rights of the students. Due to the privacy restrictions, you will need to refer to your syllabus for each class for specific information regarding the collection and the return of student work. For more information on FERPA, please go to <http://www.ed.gov/policy/gen/guid/fpco/ferpa/index.html>

Diversity and Inclusion:

Integrating a diverse set of experiences is important for a more comprehensive understanding of science. Please contact me (in person or electronically) or submit anonymous feedback **if you have any suggestions to improve the quality of the course materials**. Furthermore, I would like to create a learning environment for my students that supports a **diversity of thoughts, perspectives, and experiences, and honors your identities** (including race, gender, class, sexuality, religion, ability, etc.) To help accomplish this:

- If you have a name and/or set of pronouns that differ from those that appear in your official university records, please let me know.
- If you feel like your performance in the class is being impacted by your experiences outside of class, please don't hesitate to come and talk with me. Remember that you can also submit anonymous feedback.
- If something was said in class (by anyone) that made you feel uncomfortable, please talk to me about it.

CU Denver Fall 2025 Academic Calendar:

<https://www.ucdenver.edu/student/student-calendars/academic/fall#ft-first-eight-weeks-1>

Welcome to the course!

I am excited to have you here and confident that, with curiosity and commitment, you will succeed and find this class meaningful to both your academic and professional journey.